ZIQI GUO

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SUMMARY

ML/Data Scientist and Ph.D. candidate with 7+ years of applied research experience in machine learning, statistical modeling, and large-scale data analysis. Experienced in GenAI, LLM/RAG systems, NLP, deep learning, Bayesian modeling, and SQL-based analytical pipelines, with project experience in demand forecasting, operations research, and inventory optimization. Proven track record translating ambiguous domain problems into analytical solutions across multidisciplinary teams.

TECHNICAL SKILLS

- **Programming & Data:** Python, SQL, R, PySpark, AirFlow, Pandas, NumPy, MySQL, MongoDB, Linux, Git
- **Machine Learning:** Scikit-learn, PyTorch, TensorFlow, statistical modeling, feature engineering, neural networks, domain adaptation, random forests, XGBoost
- **GenAI / NLP:** LLMs, RAG, transformer models, sentiment analysis, SentenceTransformers, reciprocal-rank fusion, retrieval and evidence-grounded generation
- **Statistics & Optimization:** Gaussian processes, stochastic kriging, Bayesian modeling, active learning, sensitivity analysis, operations research, mixed-integer linear programming (MILP), inventory optimization, OR-Tools/SCIP
- **Data & Analytical Workflows:** ML/NLP pipelines, data preprocessing, model validation, data visualization

EXPERIENCE

Computational Image Analysis Intern Jun 2024 – Jul 2024

Banner Health – Banner Alzheimer's Institute *Phoenix, AZ*

– Built a longitudinal modeling pipeline using amyloid PET data from 990 participants across two independent Alzheimer's cohorts, combining systematic outlier detection, logistic regression for probabilistic biomarker thresholding, and non-parametric disease-progression modeling.

– Modeled a 30-year trajectory of amyloid accumulation and generated individualized biomarker-onset estimates, enabling cross-cohort analysis of disease progression from heterogeneous short-term longitudinal observations.

– Collaborated with neuroscientists and clinical researchers to translate biomarker questions into statistical analyses and communicated findings through visualizations, technical presentations, and a conference poster.

Research Assistant Aug 2018 – Present

Binghamton University *Binghamton, NY*

– Conducted applied data-science research spanning LLM/NLP systems, large-scale data mining, deep learning, transfer learning, and statistical modeling, with outputs published in ICWSM workshop, Journal of Biomedical Informatics, Scientific Reports, etc.

Generative AI, NLP & Large-Scale Data Systems *May 2021 – Present*

– Managed and prepared 1TB+ of heterogeneous data across Linux servers using Python, SQL/MySQL, and MongoDB, selecting relational or document-based storage based on data characteristics and performing extensive cleaning, deduplication, and preprocessing for downstream ML analyses.

– Built an end-to-end ML pipeline to analyze 398K+ posts from 41K+ users, fine-tuning transformer-based relation and sentiment models to distinguish adverse reactions from treatment indications and prioritize 54 high-priority candidates for downstream expert verification.

– Developed a retrieval-augmented evidence-verification system combining semantic and lexical search, rank fusion, LLM adjudication, and source-grounded reasoning across biomedical literature; AI assessments achieved 100% agreement with reviewer consensus on cases where both independent reviewers agreed.

– Analyzed 22.9M+ social-media posts across X, Instagram, Reddit, and the Epilepsy Foundation of America, developing graph-based cohort-selection algorithms to identify high-signal users from noisy population-scale data.

– Reduced network complexity to $\sim 16\%$ of original edges across three platforms ($\sim 37\%$ on X); independent human annotation validated the filtering approach, with retained Instagram users producing 85.6% relevant content versus 71.9% for non-retained users ($p < 0.01$).

– Mentored an undergraduate researcher on study design, ML workflows, database maintenance, academic writing, and presentations; secured a $5,000 mentoring grant and supported two conference manuscripts plus presentations at two conferences.

Research Assistant, Binghamton University (continued)

Supply Chain Demand Forecasting & Inventory Optimization

- Developed regularized Poisson regression forecasts for 30 store-product time series using synthetic retail data, reducing validation weighted absolute percentage error (WAPE) by 26.4% versus a seasonal-naive baseline across eight rolling weekly backtests.
- Formulated and solved a mixed-integer linear program (MILP) for inventory allocation under stock and shipping-capacity constraints, reducing base-case modeled shipping, holding, and shortage cost by 9.21% versus proportional allocation.
- Estimated safety-stock targets from weekly forecast errors and evaluated allocation across three disruption scenarios and 11 capacity settings, identifying inventory bottlenecks and store-level service risks to inform replenishment and capacity recommendations.

Domain Adaptation, Machine Learning & Medical Imaging *May 2020 – May 2022*

- Developed an unsupervised domain-adaptation framework to transfer fall-risk models across heterogeneous wearable-sensor cohorts of 171 older adults and 49 osteoporosis patients, addressing distribution shift without requiring labeled target-domain data.
- Combined covariance alignment and distribution adaptation to achieve 81% target-domain accuracy, outperforming established domain-adaptation baselines by 8–14 percentage points.
- Developed a TensorFlow-based CNN pipeline for brain-tumor classification from MRI, using synthetic-data augmentation to address limited training data; selected as a finalist in the IISE Data Analytics Competition.

Statistical Modeling & Active Learning *Aug 2018 – Aug 2019*

- Developed surrogate models integrating generalized polynomial chaos and stochastic kriging to reduce the computational cost of modeling stochastic systems.
- Designed Gaussian-process expected-improvement and batch-sampling strategies for active learning, sequential experimental design, and contour estimation.

Data Analyst Intern Aug 2018 – Jan 2019

Global Health Impact Organization *Binghamton, NY*

- Compiled and analyzed pharmaceutical research and patent data for HIV/AIDS, tuberculosis, and malaria to support development of the Global Health Impact index across three years and 216 countries.
- Collaborated with a six-person analyst team to build statistical models in R and conducted hundreds of sensitivity analyses in R and Excel to quantify uncertainty in index outputs.
- Visualized and interpreted analytical results and communicated findings through statistical reports and team presentations.

EDUCATION

Ph.D., Industrial and Systems Engineering, Binghamton University, US — Expected Dec 2026

Focus: Machine Learning, Predictive Analytics, Large-Scale Data Mining

M.S., Industrial and Systems Engineering, Binghamton University, US — May 2019

B.S., Engineering Management, China University of Petroleum, CN — May 2017

SELECTED PUBLICATIONS

1. **Guo, Z.**, Palermo, R., Hamed, A. A., & Rocha, L. "Uncovering Underreported Adverse Drug Reactions in Epilepsy Communities via Social Media Mining", Proceedings of the 19th International AAAI Conference on Web and Social Media (ICWSM-2025). https://doi.org/10.36190/2025.54.
2. **Guo, Z.**, Felag, J., Rozum, J., Correia, R. and Rocha, L. "Selection of relevant patient cohorts from social media using the metric backbone of knowledge networks". Journal of Biomedical Informatics (2025): 104847. https://doi.org/10.1016/j.jbi.2025.104847
3. **Guo, Z.**, Wu, T., Lockhart, T., Soangra, R. and Yoon, H. "Unsupervised domain adaptation for prediction of fall risk," *Scientific Reports* (2024): 3477. https://doi.org/10.1038/s41598-024-54053-5.
4. Che, Y., **Guo, Z.**, and Cheng, C. "Generalized polynomial chaos-informed efficient stochastic Kriging". *Journal of Computational Physics* (2021): 110598. https://doi.org/10.1016/j.jcp.2021.110598.

SELECTED AWARDS & CERTIFICATIONS

- **Outstanding Teaching Assistant**, Binghamton University — 2026
- **BUPNUR Grant - Mentoring Undergraduate Students Research ($5,000)**, Binghamton University — 2024
- Women Investing in Science and Health (WISH) Scholarship ($1,500), Banner Health — 2024
- **Data Analytics Competition Finalist**, 2022 International IISE Annual Conference & Expo — 2022
- **Lean Six Sigma Brown Belt Certification**, Binghamton University — 2019
- **Lean Six Sigma Green Belt Certification**, Binghamton University — 2018